

Neuroprediction of violence and criminal behavior using neuro-imaging data: From innovation to considerations for future directions

Josanne D.M. van Dongen^{a,*}, Yudith Haveman^a, Carmen S. Sergiou^b, Olivia Choy^c

^a Department of Psychology, Education and Child Studies, Erasmus University Rotterdam, the Netherlands

^b Department of Adolescent and Child Psychiatry and Psychosocial Care, Amsterdam UMC, the Netherlands

^c Department of Psychology, Nanyang Technological University, Singapore

ARTICLE INFO

Keywords:

Neuroprediction
Recidivism
Antisocial behavior

ABSTRACT

Violent conduct in society is a major health concern, and therefore one of the major aims in forensic mental healthcare is the assessment of the risk for (future) violence. The prediction of violence risk is traditionally done by using violence risk assessment tools in the form of actuarial instruments and structured professional judgments. However, research has shown that the risk assessment tools that are currently being used, are not as accurate in predicting future risk as one would hope for. Therefore, some scholars advocate a paradigm shift by incorporating a biopsychosocial model, including neurobiological measures in risk assessment models. The current review aimed to give answer to the question whether neuro-imaging measurements add in the accuracy of predicting aggression and violent (criminal) recidivism, and to discuss what next steps need to be taken to move neuro-imaging informed risk assessment forward in forensic clinical practice. The results of the current review showed that findings regarding 'neuroprediction' are mainly based on cross-sectional studies and that studies using rigorous methods for determining the incremental predictive accuracy of neural parameters are lacking. Future studies need to rely on prospective and longitudinal data that address the added value of neural measures over traditional risk factors specifically, taken different ethical and judicial considerations into account. Studies that take a personalized approach, focusing on neurobiologically informed risk assessment at the individual level to translate this into forensic clinical practice, is in our view the next step that will move this forward.

1. Introduction

Violent conduct in society is a major health concern. Violence posed by individuals with a history in forensic psychiatry is not only an economic concern, with the recent estimated annual cost of crime in the US being \$4.71–\$5.76 trillion (Anderson, 2021), but also one for public health, and social care (Senior et al., 2020). One explanation for this concern might lie in the fact that forensic psychiatric patients who are discharged from a forensic facility, are found to continue to pose a risk for violence. One meta-analysis conducted in England and Wales found violent reoffending rates of approximately 4 % per year, and rates for violent recidivism of forensic patients in the Netherlands stand at approximately 12 % within 2 years of discharge and almost 20 % after 5 years of discharge (Bogaerts et al., 2018; Fazel, Chang, et al., 2016; Fazel, Fimińska, et al., 2016). In other words, one out of five forensic

clients will re-offend again within five years after they are discharged.

The above numbers illustrate the necessity to accurately assess the risk for future violence of these individuals. One means to assess violence risk is using risk assessment tools to 'predict' who will be likely to pose a risk for future violent recidivism. Violence risk assessment tools are designed to evaluate the likelihood of someone engaging in violent behavior. These tools use various static and dynamic risk factors such as past behavior, mental health history, and situational factors to assess the risk level. They help professionals like forensic psychologists, social workers, and law enforcement to make informed decisions about interventions, treatment plans, safety measures and decisions concerning leaves or parole. Some common tools include the Violence Risk Appraisal Guide (VRAG; Harris et al., 1993), which is an actuarial tool, and the Historical Clinical Risk Management-20 version 2 (HCR-20^{V2}; Webster et al., 1997), a structured professional judgement tool.

* Corresponding author at: Department of Psychology, Education and Child Studies, Erasmus University Rotterdam, Mandeville Building T13-26, PO BOX 1738, 3000 DR, Rotterdam, the Netherlands.

E-mail address: j.d.m.vandongen@essb.eur.nl (J.D.M. van Dongen).

<https://doi.org/10.1016/j.avb.2024.102008>

Received 21 May 2024; Received in revised form 16 July 2024; Accepted 16 September 2024

Available online 19 September 2024

1359-1789/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

However, although structured tools have been widely used to assess violence risk in forensic mental health settings, there is uncertainty about the extent and quality of evidence of their predictive performance (see for instance Ogonah et al., 2023). Previous studies already showed that risk assessment tools' predictive accuracy, measured by the Area under the ROC Curve (AUC) ranges between 0.52 and 0.78 (Koh et al., 2020; Singh et al., 2011; van der Put et al., 2019), which is modest at best given the general acceptance that AUC probabilities <0.80 are of limited clinical utility (Çorbacioğlu & Aksel, 2023). That is, in terms of clinical utility of tests, the general accepted values for diagnostic accuracy of tests, also in medicine, are AUC 0.6–0.7 is poor, AUC 0.7–0.8 is fair, AUC 0.8–0.9 is good and AUC 0.9–1.0 is excellent (Borrebaeck, 2017). Therefore, one would want to aim for at least good predictive accuracy i.e. > 0.80. The receiver operating characteristic curve (ROC curve) plots a risk assessment tool's false positive rate against its true positive rate across score thresholds. The AUC measures the entire two-dimensional area underneath the entire ROC curve and is the probability that a randomly selected individual who committed a crime in the past receives a higher risk classification than a randomly selected individual who did not commit a crime in the past. The AUC is currently considered the accuracy estimate of choice in violence risk assessment when measuring predictive accuracy (Mossman, 1994; Swets et al., 2000).

Ogonah et al. (2023) also reported that most studies had a high risk of bias (49 [98 %] of 50), partly due to poor analytical approaches, and pooled AUCs for predicting violent outcomes ranged from 0.69 for both the HCR-20^{V2} (Webster et al., 1997) and VRAG (Harris et al., 1993), to 0.64 for the Static-99 (Hanson & Thornton, 2000). This shows that risk assessment tools' predictive accuracy leaves room for improvement.

One way of improving risk assessment might be by using neurobiological parameters. Although this might be practically quite a challenge, different scholars already proposed such a strategy years ago and more recently (Jansen, 2022; Kempes, 2023; Popma & Raine, 2006; van Dongen & Franken, 2019). This current review has the aim of providing an overview of previous empirical research using neurobiological, especially neuroimaging parameters to predict (violent) recidivism. Although previous research has demonstrated promising results, it is still unclear to what extent it really could impact current risk assessment practices. Also, given the focus on neurobiology, different legal and ethical concerns may arise in the implementation of such methods in forensic clinical practice. Although these topics have been discussed by different scholars (e.g. Aharoni et al., 2022; Nadelhoffer et al., 2012; Nadelhoffer & Sinnott-Armstrong, 2012; Poldrack et al., 2018; Tortora et al., 2020), currently there is no up-to date review that systematically examines the current evidence for the accuracy of neurobiological indices for the prediction of violent recidivism. Additionally, we also want to critically discuss where we would like to go with this if we want to actually implement such practices. Therefore, this current review aimed to give a systematic overview of the studies that showed predictive performance of neurobiological characteristics on aggression and/or violent behavior and recidivism and to critically discuss what next steps need to be taken to address the ethical and legal concerns of neurobiologically informed risk assessment.

1.1. Violence risk assessment in forensic clinical practice

The assessment and management of risk for violence is the core business of forensic mental healthcare. To do so, risk assessment instruments aid in the identification of individuals at risk of violence perpetration and assist in risk management. >400 risk assessment instruments designed to assess the risk of specific forms of violence and offending have been developed over the years and many have been implemented in forensic mental health around the world (Singh et al., 2014).

Different types of risk assessment tools have been designed. Actuarial risk assessment tools such as the VRAG (Harris et al., 1993), and the more recently developed OxRec (Fazel, Chang, et al., 2016; Fazel,

Fimińska, et al., 2016) estimate the likelihood of misconduct by assigning numerical values to risk factors associated with (re-)offending. These numbers are then combined using a statistical algorithm to translate an individual's total score into a probabilistic estimate of offending. The actuarial approach is an attempt to ensure that each individual is appraised using the same criteria and, in doing so, can be directly compared to others who have had the same tool administered regardless of who conducted the assessment.

Clinical approaches to risk assessment can be divided into unstructured professional judgments and structured professional judgments. Unstructured professional judgments refer to a clinician's subjective prediction of whether an individual is likely to offend or not. No pre-specified set of risk factors is used to guide the clinician's analysis of thoughts of reasoning, but the judgement relies on professional experience (Hanson, 1998). Recent reviews have suggested that this form of risk assessment has poor predictive validity (Daniels, 2005; Hanson & Morton-Bourgon, 2009), presumably because of its reliance on subjective risk estimates that lack inter-rater and test-retest reliability (Hanson & Morton-Bourgon, 2009).

To increase construct validity and reliability, the authors of risk assessment tools developed new measures that adopt an approach known as structured professional judgement (SPJ). In this approach, clinicians use empirically based risk factors to guide their predictions of an individual's risk for offending. Advocates of this approach believe that SPJ does more than simply assess risk, it provides information that can be used for risk management, including treatment planning (Douglas et al., 2013; Heilbrun, 1997). While past reviews have provided evidence that actuarial tools produce higher rates of predictive validity than instruments which rely on structured professional judgement (Hanson & Morton-Bourgon, 2007, 2009), other researchers have presented evidence that both forms of risk assessment produce equally valid predictions (Guy, 2008; Singh et al., 2011).

Current clinical guidelines recommend, or even demand using structured professional assessment tools, such as the HCR-20^{V3}, or the Historical Clinical Future – Revised (in Dutch the HKT-R; Bogaerts et al., 2018) for making decisions in court, decisions for leave and parole in forensic populations. However, despite our reliance on these tools and their widespread use, as outlined before, evidence for their predictive performance is limited. It is therefore suggested that a more holistic approach is needed, including other domains of factors that might be related to increased risk of violence. One of those domains is neurobiology. Why? Because cognition, emotion, and behavior cannot exist without our neurobiology (Gazzaniga, 2014). Although there are of course different modalities in which neurobiology can be measured, in this review we focus on the neural level, the central nervous system, and related structural and functional neuro-imaging findings related to aggression and violent re-offending.

1.2. Structural imaging findings in relation to aggression/violence

As a risk factor for violent behavior, brain abnormalities that impair social, cognitive, and affective functions may provide further insights into violent offending and thus violence risk prediction. This notion is based on a substantial body of literature showing that frontal cortical structures that are involved in inhibitory mechanisms, decision-making, and emotional processing are particularly implicated in aggressive and violent behavior as assessed in structural magnetic resonance imaging (sMRI) and diffusion tensor imaging studies (DTI). In a meta-analysis of sMRI studies that examined various frontal regions including the orbitofrontal cortex, dorsolateral prefrontal cortex, medial prefrontal cortex, and anterior cingulate cortex, prefrontal structural deficits were observed in violent individuals (Yang & Raine, 2009). Moreover, lower cortical thickness in the prefrontal cortices bilaterally have been associated with a greater frequency of aggressive behaviors in adults (Sheehan et al., 2021). These findings of prefrontal deficits have also extended to sexual violence as decreased fractional anisotropy in white

matter near the medial frontal pole indicated by DTI were found in males who had committed rape (Chen et al., 2016) and appear to be more specific to physical aggression rather than verbal aggression (Chester et al., 2017). Notably, lower prefrontal cortical volumes are not only found in healthy adults, but also in patients with psychoses who are violent (Widmayer et al., 2018). Additionally, another review and meta-analysis (Lamsma et al., 2017) found that only gray matter volumes in the parietal area were associated with aggression and violence outcomes, although these associations were spread across a large area in the parietal lobe. Additionally, the authors concluded that regional gray matter volume was not consistently associated with violent outcomes.

In sMRI studies, another brain region that shows volumetric reductions in individuals who exhibited violence and aggression in both healthy and psychiatric populations is the amygdala (e.g., Matthies et al., 2012; Widmayer et al., 2018). The amygdala-violence relationship is further supported by the finding in a prospective study that lower amygdala volumes in men predicted adult violence 3 years later (Pardini et al., 2014). In light of anatomical connections that limbic prefrontal regions have with the amygdala, it has been suggested that connectivity between these brain regions may underlie violent behavior (Rosell & Siever, 2015). Increasingly however, research has also pointed to the involvement of other brain areas. Some evidence indicates that a thinner cortex in the bilateral temporal lobe (e.g., Sheehan et al., 2021) which is implicated in perspective-taking and empathy, and increased putamen volume which is associated with reward processing (e.g., Widmayer et al., 2018), are associated with more aggressive and violent behavior. Taken together, these findings provide support for the existence of structural abnormalities in violent individuals.

1.3. Functional imaging findings in relation to aggression/violence

Consistent with structural imaging findings, meta-analytic evidence from functional magnetic resonance imaging (fMRI) studies has shown involvement of the prefrontal cortex and the amygdala in aggression. Activation in the right inferior frontal gyrus and the left amygdala were found to be enhanced in individuals who have a history of aggression during aggression-eliciting conditions, while greater activation of the left hippocampus and left parahippocampal gyrus, as well as the superior, middle, and inferior temporal gyri in aggressive individuals compared with controls suggest that other limbic areas and temporal regions may also be implicated in such behavior (Nikolic et al., 2022). In addition, significant clusters of convergent aberrant brain activation were found in the right precuneus in violent samples with high levels of trait aggression (Wong et al., 2019). The finding suggested that escalated aggression arises from abnormal activity in this regulatory brain region, which is coactivated with the lateral frontal cortex and the posterior parietal cortex within the frontoparietal network. This abnormality may disrupt the recruitment of other large-scale networks such as an adaptive cognitive control network comprising brain regions such as the midcingulate cortex, right rolandic operculum, precentral gyrus, and the postcentral gyrus, resulting in reduced abilities to generate adaptive responses. Conversely, in one activation likelihood estimation meta-analysis of studies that elicited aggression in fMRI behavioral paradigms, the right postcentral gyrus which might be involved in a common cognitive control network, was consistently involved in aggression, but not the prefrontal cortex or amygdala (Wong et al., 2019). Evidence from studies investigating brain morphology and brain activity suggests a broad network of brain regions is associated with violence. However, findings from functional neuroimaging studies are more mixed, which may be attributable in part to differences in fMRI paradigms.

Another method of investigating brain activity is using electroencephalography (EEG). Early investigations of brain differences in violent criminal offenders focused on abnormalities in EEG activity. One way of studying dysfunctions in brain structures is using resting-state electroencephalography (rsEEG). Resting-state is the state in which an

individual is awake, but not engaged in a specific task allowing to reflect the intrinsic activity of the brain (Deco et al., 2011; Greicius et al., 2003). Regarding criminal behavior and violence, previous research has shown decreased arousal in the frontal brain regions (Decety et al., 2015); demonstrating enhanced slow-wave activity of the delta (0.5–3.5 Hz) and theta band (3.5–7.5 Hz) in the brain of antisocial individuals, and decreased alpha activity in criminals (Ellingson, 1954; Mednick et al., 1981; Raine et al., 1990). Common findings in these studies investigating EEG of antisocial and violent individuals show that this excessive slow-wave activity is related to aggressive tendencies within these individuals (Brower & Price, 2001; Hill & Waterson, 1942; Moya-Albiol, 2004; Scarpa & Raine, 1997). Furthermore, besides the higher power of activity in slow waves, the reduced power in the alpha frequency band (8–12 Hz) also seems one of the more robust findings in individuals with antisocial and behavioral disorders (Calzada-Reyes et al., 2013; Convit et al., 1991; Keune et al., 2012). Given the impaired neural activity of violent offenders in the alpha frequency band, it is plausible that impaired brain communication reflects one of the neuroanatomical key factors to antisocial and violent individuals (Motzkin et al., 2011; Yang & Raine, 2009).

Early research has unveiled a correlation between heightened slow wave EEG activity during adolescence and the subsequent onset of antisocial behavior in adulthood, as evidenced by official criminal convictions (Raine et al., 1990). Additionally, Gatzke-Kopp et al. (2001) found increase delta activity in temporal lobes in murderers. Theoretical frameworks positing links between augmented slow wave EEG patterns and various forms of criminal behavior, including violent offences, have primarily centered on two key concepts. First, cortical immaturity has been proposed as a factor leading to impaired inhibitory control (Volavka, 1999). Second, the notion of cortical under-arousal has been suggested, which may predispose individuals to seek compensatory stimulation (Raine et al., 1990). Therefore, these findings indicate that violent offenders show abnormalities in frontal regions, including enhanced power in the theta frequency band and decreased alpha power and impaired connectivity. These findings are supported by more recent studies of Tillem et al. (2016, 2018) that found a higher theta coherence in male inmates with psychopathic traits and alterations in the efficiency of neural communication. Furthermore, Lindberg et al. (2005) found an increase of theta and beta powers in the occipital regions, and a decrease alpha frequency in the EEG. Lastly, the previous described alterations in the EEG of violent offenders with antisocial behavior were demonstrated in the study of Reyes and Amador (2009), which found decreased alpha and increased slow-wave activity.

In addition to this EEG slow wave research, increasing evidence suggests asymmetrical involvement of the frontal hemispheres in aggressive behavior (for review see Harmon-Jones, 2003). Particularly, the left hemisphere has been found dominant in aggressive behavior (Carver & Harmon-Jones, 2009; Peterson et al., 2008; Schutter et al., 2008), and studies link increasing activity with higher levels of experienced anger and aggressive behavior (Peterson et al., 2008). Furthermore, the studies of Hofman and Schutter (2009, 2012) have demonstrated that asymmetrical activity in the beta frequency band specifically is found to be related to the development and expression of aggressive behavior (Hofman & Schutter, 2012) and in trait aggression in healthy individuals.

Another way of investigating the EEG in regards to aggressive behavior is through analyzing event-related potentials (ERPs), measuring the electrophysiological response of the brain to an internal or environmental event. The existing literature on ERP responses in aggressive individuals is less extensive compared to EEG studies, yet various ERP components have been linked to aggression (see Patrick, 2008 for a review; Sergiou et al., 2023). Among these, the most consistent association lies with the reduced amplitude of the P300 response component, particularly evident in impulsive forms of aggression (Barratt et al., 1997; Wang et al., 2012). Additionally, reduced P300 amplitude has been observed in individuals diagnosed

with antisocial personality disorder, alongside other impulse control issues such as alcohol and drug dependence, nicotine dependence, child conduct disorder, and attention deficit hyperactivity disorder, and antisocial personality traits (Bauer et al., 1994; Gao & Raine, 2009; Iacono et al., 2003). This reduction may reflect a shared aspect among these disorders rather than a unique characteristic of any one of them, possibly related to a broader “externalizing” factor (Patrick et al., 2001).

Most studies on aggressive individuals using EEG and ERP measures are marker studies focusing on brain activity at rest or during simple tasks. The fine-grained information available from EEG/ERP measurement in time and frequency domains alongside advancements in source localization methods, holds promise to deepen our understanding into aggressive individuals using electrocortical activity in future studies (Patrick, 2008; Sergiou et al., 2023).

A third tool, functional near-infrared spectroscopy (fNIRS), has been regarded as a complementary technique to fMRI and EEG to assess brain activity (Quaresima & Ferrari, 2019). One study using fNIRS reported that during an aggression task, abnormal correlations between oxygenated and deoxygenated hemoglobin concentration changes within the prefrontal cortical network were observed in psychiatric patients with addiction (Asaoka et al., 2021). Although the altered prefrontal cortical activity patterns were not observed in healthy controls performing the aggression task and could be argued to be attributed to other aspects of task performance rather than aggressive behavior specifically, the results provide some preliminary evidence for the involvement of the prefrontal cortex in aggression using a different neuroimaging technique. The paucity of research using fNIRS in elucidating the neural basis of aggressive and violent behavior suggest these findings require replication.

1.4. Current review

Current violence risk assessment and management are mainly focused on psychological and social risk factors in predicting future risk but are at the same time in need of a paradigm shift. Previous research has shown that a biopsychosocial model of antisocial behavior would be a good candidate in making this shift. That is, (persistent) antisocial behavior can be seen as a neurodevelopmental disorder (e.g. Raine, 2018; van Goozen et al., 2022), and past research has found multiple neurobiological underpinnings that may explain persistent antisocial behavior, including aggression and violent behavior.

One of the aims of this review is to discuss to what extent neuroscientific correlates, that is, neurophysiological measures using neuroimaging can contribute to a better prediction of (recurrence of) violence - often referred to as “neuroprediction”. Neuroprediction can be of great importance for forensic clinical practice, in which estimating recidivism risk plays a crucial role, but where commonly used risk assessment tools prove to be far from perfect. Neural predictors, or biomarkers, are objective biological measures that are either diagnostic biomarkers that index a biological process associated with health or disease, or in this case violence or antisocial behavior, or predictive biomarkers that reflect a process associated with the therapeutic response (i.e. reduction in violence risk) and are used in clinical stratification (Abi-Dargham & Horga, 2016). The use of neural predictor or biomarkers are in line with the precision-medicine initiative, a movement now also more and more present in psychiatry.

Recent scientific studies tentatively confirm the potential value of the neurobiological information in this regard (e.g., Aharoni et al., 2013; de Ruigh et al., 2021; Zijlmans et al., 2021). At the same time, the use of neurotechnology in criminal law also raises legal and ethical questions, such as about bias and transparency of the data used, autonomy, mental privacy, freedom of thought and mental integrity of the forensic client subjected to neurotechnology (e.g. Aharoni et al., 2022; Meynen et al., 2023; Tesink et al., 2024).

It is precisely these previous and current insights from neuroprediction of (violent) recidivism that we want to review, and the

related legal-ethical questions that go with that to discuss the future ahead for forensic mental healthcare, to come to a future next step.

One might wonder why the focus of this review is on neuroimaging findings, and not on other additional neurobiological correlates. This is because 1) we aimed to keep it concise, given the fact that we will also discuss the ethical and legal implications specific to this domain, and 2) because neuro-imaging findings might give us insight in future neurobiologically informed treatments using, for instance, neuromodulation techniques to modulate (functional) brain networks (To et al., 2018) to reduce aggressive behavior in forensic patients and violent offenders (Molero-Chamizo et al., 2019; Sergiou et al., 2022; Sergiou et al., 2023), and even healthy controls (see Casula et al., 2023 for a review), and 3) individual differences in neurobiology, including brain networks affect treatment response and are therefore of great importance to assess in the risk assessment phase before managing this risk in the treatment of forensic patients (i.e. precision medicine and personalized treatments; Hampel et al., 2023).

Thus, although other modalities of neurobiology such as heart rate and skin conductance are regarded as potential relevant neurobiological parameters for neurobiologically informed risk assessment, also because of the potential more mobile and cost-efficient measurement of those parameters compared to neuro-imaging measures (Jansen, 2022), those measures are peripheral physiological measures of autonomic nervous system activity, and not central neural activity. Therefore, neural measures, as assessed with structural and functional neuro-imaging devices are the correlates we will focus on in this review. Additionally, we aimed to review the literature on the added value of these neuroimaging measures over more traditional psychological and social factors used in predictive future aggressive, violent and or criminal behavior.

2. Methods

2.1. Search strategy

This systematic literature review was conducted in accordance with the PRISMA guidelines (Page et al., 2021) (See Fig. 1). The literature search was conducted in PubMed and Web of Science (WoS) between August and November 2023 by two authors (YH and CS). The search was conducted based on a research question originating from the PICO model (Page et al., 2021). It involved individuals with violent/criminal behavior (P) and examined whether the use of neuro-imaging measures (I) adds value to the commonly used psychosocial measures (C) in predicting future violent/criminal behavior (O). Based on the PICO model, the search term was constructed from the following components: (1) neuro-imaging measures, (2) violent and forensic behavior, and (3) prediction.

Search terms related to neuro-imaging measures, violent behavior and prediction ((Crim* OR Forensic OR Aggression OR Impulsivity OR Violence OR Offender* OR ((criminal OR antisocial OR violent OR impulsive OR aggressive OR conduct OR externaliz*) AND (behavior* OR disorder*))) AND (Neuropredict* OR ((predict* OR “risk assessment” OR “violence risk” OR Recidivism OR Reoffend* OR Rearrest) AND (neuro* OR brain* OR neurobiology* OR neuroscience*))) NOT (neuroticism OR cancer OR tumor*)) were used.

Synonyms for each component were devised and carefully weighed based on the number of hits. Besides, relevant MeSH terms (medical subject headings) were added in the PubMed search string, and reference lists of relevant articles were screened for other potential synonyms. All synonyms and MeSH terms were combined with an ‘AND’ operator within each of the three components and separated by an ‘OR’ operator between the three components. Results were limited to human studies written between January 2008 and November 2023, and language of publication restricted to English. All results were downloaded to an excel sheet, and duplicates were removed.

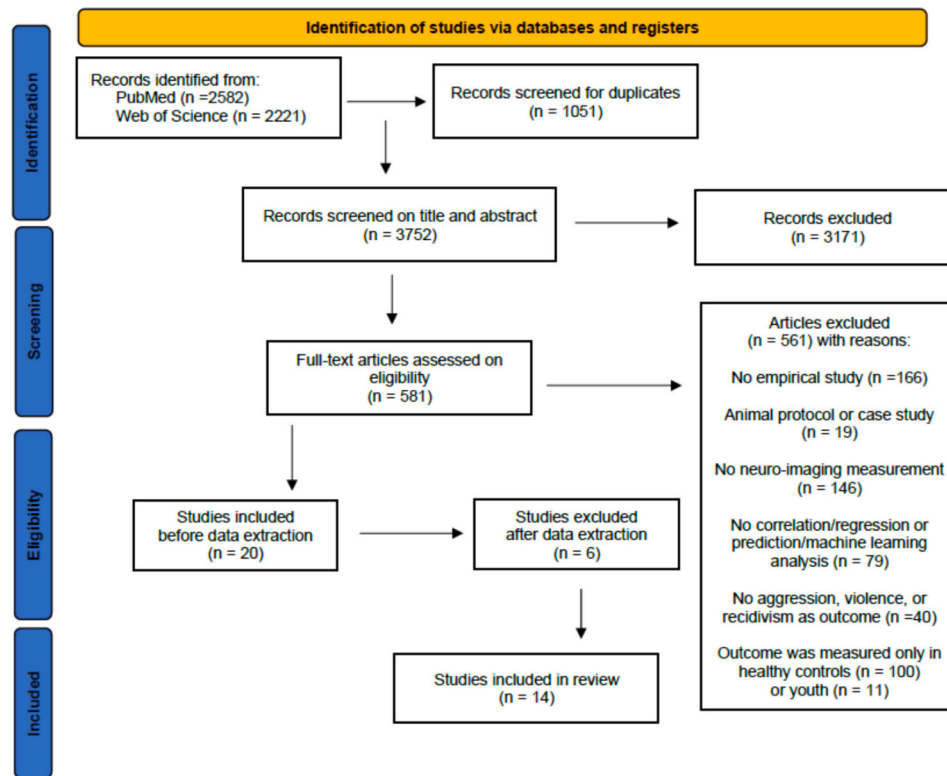


Fig. 1. PRISMA flow chart of the search strategy and the selection of studies.

2.2. Eligibility criteria

Articles' eligibility criteria to be included in the review were that they had to contain published data, and the use of structural or functional neuro-imaging measures to study aggressive/violent (criminal) behavior in a 'predictive' manner. That is, studies were included when a prospective longitudinal design was used, and data were analyzed using prediction models. Following the initial search and removal of duplicates, authors (YH, CS) screened the articles by title/abstract. An article had to generally align with the PICO terms where some flexibility was in place since this screening was carried out parallel to each other. Subsequently, a second selection round took place, where the content of the articles was assessed more rigorously. A standardized data collection form was developed and used within excel to extract information about the study design, reported measured predictors, outcomes, and the analyses used. Full texts of the included studies were divided among the reviewers (YH, CS), but were checked very regularly on sampling basis, and any ambiguities were discussed. From here, in consultation with the other authors (JD, OC), a selection protocol for inclusion was created. These criteria were listed precisely and concrete.

Studies were excluded when they did not entail an empirical study (meaning that reviews, case-studies and protocol-studies were excluded) in a human sample. Besides, there must be at least one neuro-imaging variable investigated and the statistical analysis must include analyses using some kind of prediction model. Because at initial inclusion, the number of studies that used prediction models as analyses appeared very small, it was decided to also include studies that used correlational/regression design to 'predict' aggression or violent (criminal) behavior. Next, together with the other authors (JD, OC) we updated the protocol with some stricter and more precise inclusion criteria. Articles including only healthy subjects and/or children were excluded unless a longitudinal design was used to predict outcomes in adulthood. Additionally, trait-like aggressive, antisocial, or externalizing behavioral variables as outcomes of the prediction were excluded, thereby focusing merely on (criminal) acts of aggression and violence, and not on psychopathic

traits for instance. All the included articles were carefully read and checked by both authors (YH, CS). During the entire process, consistency was monitored through regular meetings between YH and CS.

2.3. Data collection process

As part of the inclusion process, a data collection form was created including sample (size, and a categorized basic description of clinical characteristics), predictors (categorized using different brain scans, psychological or biological variables), outcome variables (as categorized by aggressive, violent or criminal behavior and recidivism) and the type of analysis (prediction, regression, prevalence, differences or relations between groups). After the inclusion of articles based on this data collection form, the included articles were further analyzed in detail in which the predictor and outcome measures (including follow-up term) were specified, as well as the statistical analysis. Besides, the significant predictors (with *p*-values) and the results of the regression/prediction analyses were extracted. Here, the focus was primarily on the added value of neuro-imaging measures compared to currently used prediction models and risk factors based on more traditional social, environmental and/or psychological variables. Where possible, Area Under the Curve (AUC) results were extracted. If the AUC was only available for neuro-imaging measures, without its comparison with psychological models, it was mentioned as well. Otherwise, the regression/prediction outcome including a measure of explained variance was extracted. Data extraction was carried out by YH, checked by CS, and discussed with the other authors.

3. Results

The literature search resulted in fourteen empirical articles concerning the prediction of aggression, violence or offending behavior or groups of offenders compared to non-offenders or aggression groups, either by studying it cross-sectionally or in a longitudinal/prospective design (see Table 1).

Table 1
Main characteristics and outcomes of the included studies ($n = 14$).^a

Cross-sectional studies: structural brain-imaging ($n = 5$)						
Article	Sample	Measured variables	Outcomes	Analysis method	Significant/important predictors	Accuracy prediction
Gou et al. (2021)	75 schizophrenia patients (42 with violent behavior, 33 not)	Brain: structural MRI and rs-fMRI Crime/violence: MOAS, PCL-SV, HCR-20 Other: BPRS, BIS-11	Presence of violent behavior	LASSO, SVM and LOOCV	Frontal gyri, striatum system, (anterior + dendate) cingulate gyrus, parietal lobe, temporal pole and superior longitudinal fasciculus ($p < 0.05$)	The model consisting of sociodemographic and clinical data achieves an accuracy of 80 % (AUC = 0.91). Adding the three neuro-imaging modalities increases the prediction accuracy to 91 % (AUC = 0.95)
Lett et al. (2018)	73 pedophiles with child sexual offense, 77 pedophiles without, 133 HC	Brain: structural MRI Crime/violence: SIS, and SES2 Other: SCID axis I&II, sexual interest, IQ	Presence of child sexual offense behavior	Regression analysis	Left hemisphere mean cluster surface area explains 30 % of the number of offences against children, the right hemisphere mean cluster surface area 21 %. There seems to be a mediation with IQ (Sobel $Z = 3.18$, $p = 0.0015$) The best classifier included bilateral fractional anisotropy in the ACC (CVR's between 6.65 and 36.76), diffusivity in the left amygdala (CVR = -13.16), and structural prefrontal cortex-amygdala connectivity in both hemispheres (CVR = 29.05 and CVR = 21.41)	No data available on the added value of neurological data compared to clinical variables.
Popovic et al. (2023)	14 pedophilic offenders and 63 HC	Brain: structural MRI Crime/violence: PCL-R, SVR-20, MSI Other: SCID-IV, verbal premorbid intelligence,	Presence of pedophile child sexual offense behavior	SVM with cross-validation ratio (CVR)	Lower left dorsomedial PFC, extending into the dorsal ACC, gray matter volume indicated a higher risk for re-offending (Spearman's $\rho = -0.408$; $P = 0.001$)	The classifier correctly identified 9 of 14 PO individuals (sensitivity = 64.3 %) and 13 of 15 HC individuals (specificity = 86.7 %), with an AUC of 0.69.
Schiffer et al. (2017)	58 pedophiles with child sexual offense, 60 without and 101 HC	Brain: structural MRI Crime/violence: SSPI-2, SIS/SES Other: SCID axis I&II, IQ, BIS-11, SPFF-IRI, Kinsey scale, 2D:4D digit ratio, viewing reaction time task	Risk of re-offending using SSPI-2	ANCOVA, pearson correlation, regression analysis	Reductions in thickness of the right banks of the superior temporal sulcus ($p < 0.001$), inferior parietal thickness ($p = 0.002$) and left frontal pole ($p < 0.001$) volume reached significance and were input for the prediction models	No data available on the added value of neurological data compared to clinical variables.
Yu et al. (2022)	57 patients with schizophrenia (30 with violent behavior, 27 without)	Brain: structural MRI Crime/violence: MOAS Other: education, marital status	Presence of violent behavior	Different machine learning algorithms		Among seven predictive models, SVM was identified as the optimal algorithm for violence in SCZ with a balanced accuracy of 0.8231 and an AUC of 0.8410
Cross-sectional studies: functional brain-imaging ($n = 3$)						
Article	Sample	Measured variables	Outcomes	Analysis method	Significant/important predictors	Accuracy prediction
Amaoui et al. (2023)	26 perpetrators, 29 other offenders and 29 without criminal record	Brain: emotional regulation task during fMRI Crime/violence: Conflict tactic scale 2 Other: DERS, ERQ and Risk of serious couple violence	Ability to down-regulate emotions to partner violence	Parametric Empirical Bayes framework with LOOCV	Connections between inferior, medial and middle frontal gyrus, the superior and middle temporal gyrus and the supramarginal gyrus ($0.01 < p < 0.44$)	No data available on the added value of neurological data compared to clinical variables.
Athanassiou et al. (2022)	62 schizophrenia patients and 25 healthy controls	Brain: rs-fMRI Crime/violence: MACVI Other: PANSS, SES	Presence of violent behavior	Pearson's correlations and regression analyses	Connectivity between the ACC and the Left OFC ($p < 0.001$)	The connectivity between the ACC and left OFC significantly predicted the number of past violent behaviors in schizophrenia patients (Wald $\chi^2 = 8.32$; $R^2 = 0.17$, $p < 0.05$)
Ter Harmse et al. (2022)	118 delinquent young adults and 25 HC	Brain: passive picture viewing task during EEG Crime/violence: RPQ Other: ECG and SCL	Proactive and reactive aggression	Regression analyses	None of the neurobiological measures significantly contributed to the explanation of proactive nor reactive aggression.	No data available on the added value of neurological data compared to clinical variables.
Longitudinal studies: structural brain-imaging ($n = 1$)						
Article	Sample	Measured variables	Outcomes	Analysis method	Significant/important predictors	Accuracy prediction
Kiehl et al. (2018)	1332 criminal offenders plus the sample of Aharoni et al. [2013]	Brain: structural MRI (brain age) Crime/violence: PCL-R Other: IQ, KSADS/SCID-I	Recidivism data (no fixed FU term)	Cox proportional hazards regressions	ACC activity ($p = 0.001$), brain age ($p = 0.035$), temporal pole gray matter density ($p = 0.005$), PCL-R F2 ($p = 0.008$)	A similar model as Aharoni et al. [2013; 2014] including ACC activity gave likewise results: Wald $\chi^2 (8) = 17.04$ ($p = 0.030$). Adding brain components and relative age increased the

(continued on next page)

Table 1 (continued)

Cross-sectional studies: structural brain-imaging (n = 5)						
Article	Sample	Measured variables	Outcomes	Analysis method	Significant/important predictors	Accuracy prediction
		plus data from Aharoni et al. [2013]				prediction model: Wald χ^2 (13) = 32.2 p = 0.001), explaining 16 % more variance.
Longitudinal studies: functional brain-imaging (n = 5)						
Article	Sample	Measured variables	Outcomes	Analysis method	Significant/important predictors	Accuracy prediction
Aharoni et al. (2013, 2014)	96 criminal offenders vs 102 healthy adults	Brain: Go/no-go task during fMRI Crime/violence: PCL-R, age at release Other: IQ, drug and alcohol use	Recidivism data after 6/12/24/36 months	Cox proportional hazards regression with AUC calculation	Any crime: ACC activity ($P < 0.001$) Non-violent crime: age at release ($P < 0.05$), PCL-R factor antisocial behavior ($P < 0.01$), ACC activity ($P < 0.001$)	Any crime: ACC activity increases the accuracy of the prediction from AUC = 0.612 to AUC = 0.68. The model remains accurate during FU. Non violent crime: AUC increases from 0.68 to 0.757 after adding ACC activity, which remains strong in the FU period.
Delfin et al. (2019)	44 forensic psychiatric patients	Brain: SPECT Crime/violence: forensic psychiatric investigation protocol, PCL-SV Other: SCID-I and SCID-II	Recidivism data (no fixed FU term)	Random forest classification with AUC calculation	Parietal cerebral bloodflow (right: $p < 0.001$, left: $p = 0.036$), cerebellum (right: $p = 0.049$) cluster B personality disorder ($p = 0.008$) and age at first crime ($p = 0.001$)	Predictive performance of the baseline model was modest with an AUC of 0.69. The accuracy increased after adding SPECT data across all metrics, achieving an AUC of 0.81
Steele et al. (2015)	Subsample from Aharoni et al. [2013] with 45 criminal offenders	Brain: Go/no-go task during fMRI and EEG Crime/violence: PCL-R, age at release Other: IQ and lifetime prevalence of drug and alcohol abuse/dependence	Recidivism data (no fixed FU term)	Cox proportional hazard regression, logistic regression and SVM	The logistic regression with the highest predictive effect for rearrests found a principal component corresponding with error positivity as only significant predictor ($\beta = 2.40$, $p = 0.038$)	SVM showed that a model only including ACC activity and PC's extracted from ERP measures reached the highest prediction accuracy (78.05 %). A model based on only psychological variables achieved an accuracy of 60.98 %, using both neurological and psychological predictors lowered the accuracy of the prediction as well (73.17 %). Demographical, clinical and behavioral traits had moderate predictive power for overall and for serious recidivism (C-index over 30 months was 0.68 and 0.75, respectively). Neurobiological measures significantly improved predictive power (C30 = 0.72 for overall recidivism and C30 = 0.80 for serious recidivism).
Zijlmans et al. (2021)	127 delinquent young adults aged 18–27 years	Brain: go/no-go task and Flanker- task during ECG, fMRI and EEG Crime/violence: previous delinquency, RPAQ, YPI Other: IQ, previous drug use; ASR	Recidivism data after 6/12/18/24/30 months	Cox proportional hazards regression with AUC calculation	General offences: Morrocan ($p = 0.043$), past offences ($p < 0.001$) and error-related negativity ($p = 0.008$) Serious offences: past offences ($p < 0.001$), psychopathy ($p = 0.049$) and heart rate ($p = 0.023$)	

* Note: MRI = magnetic resonance imaging, rs-fMRI = resting-state functional MRI, EEG = electro-encephalography, ECG = electrocardiography, SPECT = single-photon emission computed tomography, SCL = skin conductance level, RPAQ = reactive proactive aggression questionnaire, YPI = youth psychopathic traits inventory, PCL-R = psychopathy checklist revised, PCL-SV = psychopathy checklist short version, MACVI = MacArthur community violence instrument, MOAS = modified overt aggression questionnaire, SSPI = screening scale for pedophilic interests, SIS/SES = sexual inhibition and excitation scale, SVR-20 = sexual violence risk - 20, MSI = McLean screening instrument, HCR-20 = historical clinical risk management - 20, IQ = intelligence coefficient, ASR = adult self-report, SCID = structural clinical interview for DSM-5, KSADS = Kiddie schedule for affective disorders and schizophrenia, DERS = difficulties in emotion regulation scale, ERQ = emotion regulation questionnaire, PANSS = positive and negative syndrome scale, BIS-11 = Barratt impulsiveness scale, SPF-IRI = Saarbrucker personality questionnaire (German version of the interpersonal reactivity index), BPRS = brief psychiatric rating scale, FU = follow-up, SCZ = schizophrenia spectrum disorder, HC = healthy control, PO = pedophilic offender, AUC = area under the curve, SVM = support vector machine, PC = principal component, ERP = event-related potentials, LOOCV = leave one out cross validation, CVR = cross-validation ratio, LASSO = least absolute shrinkage and selection operator, ACC = anterior cingulate cortex, OFC = orbital frontal cortex, PFC = prefrontal cortex.

The results are structured in different paragraphs, discussed by sections concerning studies using a cross-sectional design and divided by structural and functional neuroimaging techniques used for the neuroimaging assessment of neural parameters.

3.1. Cross-sectional studies

More than half of the studies that were found based on the search results ($n = 8$) are studies that used a cross-sectional design. Notably, the studies involved 'predictions', but mostly they predicted whether an individual could be classified for group membership, i.e. offender versus non-offender for instance, based on neuro-imaging measures.

3.1.1. Structural brain-imaging studies

Five of the cross-sectional studies used structural neuro-imaging methods, of which three were on sexual violence. [Schiffer and colleagues \(2017\)](#) were the first to distinguish the brain structural anomalies associated with pedophilia from those specifically associated with child sexual offences (CSO) in pedophilic men. The researchers analyzed the gray matter (GM) volume of 58 pedophiles with a history of CSO, 60 pedophiles without any history of CSO and 101 non-pedophilic, non-offending controls while controlling also for the effects of age, education level, verbal IQ, sexual orientation, drug misuse/dependence, other Axis I or II disorders or general criminality. Although there were no differences in the relative GM volume of the brain specifically associated with pedophilia, they did find a significant and CSO-related pattern of above vs below the 'normal' GM volume in the right temporal pole (TP), with

non-offending pedophiles exhibiting larger volumes than offending pedophiles.

Interestingly, the GM volume in the right TP was positively correlated with affective empathy skills and the propensity to inhibit sexual behavior due to potentially negative consequences of sex in non-offending pedophiles. Moreover, regression analysis revealed that the lower GM volume of the dorsomedial prefrontal cortex (i.e. ACC) was associated with a higher risk of re-offending in pedophilic child molesters. The authors concluded that their findings contribute to the existing literature on the deficient inhibitory control in pedophilic CSOs.

Lett et al. (2018) found similar results by predicting group membership of pedophiles with child sexual offences ($n = 73$), and pedophiles without such offences ($n = 77$), and non-pedophilic controls ($n = 133$) based on multimodal structural neuroimaging measures (i.e., cortical thickness, surface area, and white matter fractional anisotropy), as well as IQ. They found that sex offenders with pedophilia have more distinct variations in the brain compared to those individuals with pedophilia without an offense history, especially with regards to reductions of cortical thickness, surface area, and white matter fractional anisotropy. Moreover, they found that IQ mediated the relation between cortical surface area and sexual offense behavior by pedophiles, most profoundly in the left prefrontal cortex and the right superior frontal cortex. They concluded that this supports the idea that those offenders with pedophilia are marked by neurodevelopmental abnormalities as reflected by the lower IQ and smaller brain areas.

In the study by Popovic et al. (2023), machine learning was used to identify offenders with pedophilia based on structural neuroimaging data (DTI) in 14 offenders and 15 matched healthy controls (HC), and subsequently assessed model specificity in an external cohort of 53 HC individuals. In this study, they did not include a group of non-offenders with pedophilia. They used regions of interest (ROI) based on previous studies, including the prefrontal cortex, anterior cingulate cortex, the amygdala and the corpus callosum. The classifier discriminated offenders from non-offenders with an AUC of 0.69 (sensitivity = 64.3 %, specificity = 86.7 %) and found predictive brain pattern contained bilateral fractional anisotropy in the anterior cingulate cortex, diffusivity in the left amygdala, and structural prefrontal cortex-amygdala connectivity in both hemispheres. Note however, that the model was especially good in correctly identifying the individuals who were predicted as being non-offenders and were indeed non-offenders. The predictive accuracy was found to be poor. Additionally, in this study, no comparison was made between models including more traditional risk factors versus models including neuro-imaging data.

The next two papers were conducted to study violence in patients with schizophrenia. Gou and colleagues (2021) build an effective model to predict which individual would belong to a group of violent offender patients with schizophrenia ($n = 42$), or a non-offender patient group with schizophrenia ($n = 33$) using machine learning and both structural and functional MRI. The model with four more traditional risk variables, i.e., hostility-suspicion, psychopathy, the overall score of violence risk, and low educational level, was able to distinguish violent offenders from non-offenders with an AUC of 0.91, sensitivity of 75.76 %, and specificity of 83.33 %. The final model, which combined multimodal data (i.e., the sociodemographic, clinical, and a combination of sMRI and fMRI data), achieved better accuracy than other fused models with a sensitivity of 90.91 %, specificity of 90.48 %, and AUC of 0.95. The most discriminative neuroimaging features involved the prefrontal-temporal cognitive circuit and striatum reward system, indicating that dysfunction in cortical-subcortical circuits might be associated with high risk of violence in patients with schizophrenia.

Yu et al. (2022) aimed to add to the outcomes of Gou and colleagues (2021) by including sMRI, and used different machine learning algorithms to predict individuals group membership for either patient with schizophrenia who had exhibited violent behavior in the past ($n = 30$) and patients without a history of violence ($n = 27$) based on sMRI including gray matter volume, cortical surface area, and cortical

thickness between.

They found that among the seven machine learning techniques, Support Vector Machine did best in differentiating patients with violence from those without violence, with an AUC of 0.84, sensitivity of 0.80 and specificity of 0.85. Patients with schizophrenia who had a history of violence displayed reduced whole brain volume and cortical thickness and volume in several brain regions, including left frontal pole, right banks of the superior temporal sulcus and inferior parietal regions. The authors concluded that their findings are in line with previous studies (Antonova et al., 2004; Puri et al., 2008), and suggest the possibility that certain general cognitive impairments associated with whole brain volume reduction are involved in violence.

In sum, prediction models in structural imaging studies were most accurate when both more traditional socio-demographic and clinical risk factors were combined with neuro-imaging data. Additional accuracy of including neuro-imaging data reached significance with on average, 10 % increased predictive value. Importantly however, models seem to be more accurate in the prediction of non-offenders who will not offend (i.e. specificity) than those offenders who will offend (i.e. sensitivity), which is consistent with findings from traditional risk assessment's predictive accuracy (Ogonah et al., 2023).

3.1.2. Functional brain-imaging studies

Three of the cross-sectional studies used functional imaging techniques to predict which group would pose a risk for violence and/or recidivism. In the study by Ter Harmse and colleagues (2022), different neurobiological parameters were used, including electrophysiological responses (i.e., P3, late positive potential, mu suppression) to images depicting victims of an aggressive act, to see how emotional or empathic processing on a neural level would relate or predict self-reported proactive and reactive aggressive behavior. In that study, they included a sample of 118 delinquent young adults and 25 controls (all male, aged 18–27), and found that, although the delinquent young adults showed diminished P3 amplitude towards victim of aggression images, using regression analyses, none of the electrophysiological indices were related to either form of aggression.

These null findings might however be explained by the fact that was a large difference in group size (hence leading to a power problem) and the fact that the validity of the passive viewing task used in the study did not seem to be proven (although task validity was supported in other studies (i.e., Sergiou et al., 2022; van Dongen et al., 2018).

The studies by Athanassiou et al. (2022) and Amaoui et al. (2023) both looked at neural connectivity associated with violent behavior, the former in schizophrenia, and the latter in perpetrators of intimate partner violence against women. Athanassiou et al. (2022) used rs-fMRI to examine differences in neural connectivity between patients with schizophrenia ($n = 62$) and healthy controls ($n = 25$). More important for this review, however, was that they also looked at the association between connectivity and number of violent behaviors in patients (measured with the MacArthur Community Violence Instrument; Steadman & Silver, 2000). They found that specifically, an increase in the connectivity between the ACC and left OFC was associated with more violent behaviors.

This increase in functional connectivity in frontal areas in perpetrators was also found by Amaoui et al. (2023). In their study, rs-fMRI was used to characterize the network dynamics underlying reappraisal. Spectral dynamic causal modelling was performed to examine the connectivity within a predefined reappraisal-related brain network. 26 men convicted for intimate partner violence were compared to 29 men convicted of other crimes and 29 male non-offenders. Results showed that compared to non-offenders, both convicted groups exhibited increased functional connectivity within prefrontal areas, enhanced connectivity from prefrontal to temporoparietal regions and decreased connectivity in the opposite direction. Perpetrators of intimate partner violence showed increased connectivity from temporoparietal to prefrontal regions and, increased connectivity from the supplementary

motor area to frontal areas compared to other offenders. Moreover, connections involving the dorsolateral prefrontal cortex were found to be potential predictors of the ability to down-regulate emotions during an emotion regulation task in perpetrators of intimate partner violence. Especially the higher the effective connectivity within the male perpetrator group, most notably the IFG and DLPFC connections to temporoparietal areas, the harder their ability to down-regulate emotions when facing intimate partner violence related stimuli. This finding supports the hypothesis of an enhanced brain network associated with emotion regulation difficulties, and hence violent behavior as a reaction to this.

In sum, compared to the structural imaging studies, two of these functional neuro-imaging studies examined neural connectivity measures as markers for violence and showed increased connectivity in especially the prefrontal cortex was related to violent behavior. However, none of the functional studies studied the added predictive accuracy of neural measures to predict violence or recidivism over and above more traditional risk factors. Together with the fact that these studies were cross-sectional in nature, this hinders inferences about using neural markers as predictors for future (violent) recidivism and points to the need for more research on this topic. Our next review of the studies with prospective/longitudinal design may shed better light on this.

3.2. Prospective/longitudinal studies

Six studies were found to be longitudinal and used a prospective design. All dealt with the prediction of future recidivism.

3.2.1. Structural brain-imaging studies

Only one study, by Kiehl and colleagues (2018) looked at structural neuroimaging parameters to predict rearrest in male offenders. In that study, the researchers first developed a brain-age model that predicted chronological age based on sMRI data from incarcerated males ($n = 1332$). They then tested the model's ability to predict recidivism in a new sample of offenders ($n = 93$), of which they had longitudinal follow-up data. The average follow-up period was 21.96 months (range: 1.51 to 49.55 months; see (Aharoni et al., 2013), and the application sample used was the same sample as that used in the studies by Aharoni et al. (2013) and Aharoni et al. (2014). Neural age measures identified in their first experiment were subsequently reduced for a prediction model using supervised machine learning algorithms (i.e., Support Vector Machines). Using this method, seven neural components were identified to be useful in predicting rearrest (versus those offenders that were not rearrested).

These neural age measures identified in this feature selection were then used.

In Cox proportional hazards regression predicting 'time to rearrest' and different models were compared. They found that brain-age measures accounted for almost four times the variance in the risk for rearrest than did chronological age, and incrementally added to risk prediction that included other variables (e.g. psychopathic traits, drug- and alcohol dependence, neural measures used in Aharoni et al., 2013) in the model. Moreover, when neural age was included in the model with chronological age, chronological age was not necessary in predicting rearrest. Specifically, reduced gray matter in bilateral anterior/lateral temporal lobes, amygdala, and inferior/orbital frontal cortex were found to be helpful in predicting rearrest.

3.2.2. Functional brain-imaging studies

Most seminal when concerned with neuroprediction of recidivism are the articles of Kiehl's group (i.e., Aharoni et al., 2013, 2014; Steele et al., 2015), which are based on the same offender sample. Although we therefore need to interpret the outcomes with caution, the articles are of great value when it comes to neuroprediction.

Aharoni and colleagues (2013) were among the first to carry out neuroprediction to predict future rearrest using fMRI measures. In a sample ($n = 96$) of male offenders, they used different psychological and behavioral assessment measures and an fMRI-based go/no-go (GNG)

impulse control task to measure ACC activity in impulsive control to test whether this would add to the prediction of future rearrest. Using Cox regression models, it was found that ACC functional activity during inhibitory control (i.e., lower activity meaning less inhibitory control and thus more commission errors) demonstrated incremental predictive validity independent of other known risk factors and the GNG commission error rate, suggesting a possible predictive advantage over some behavioral and personality risk factors such as age, psychopathic traits and alcohol- and drug dependence.

Aharoni et al. (2014) published a follow-up of this study using discrimination and calibration techniques to test the accuracy of these predictions relative to more traditional models, because Cox Proportional Hazards Regression on a single dataset does not assess a model's degree of accuracy at predicting event outcomes. In other words, it does not tell us how well the model explains between those who were rearrested and those who were not, known as *discrimination*. Also, the outcomes of Aharoni et al. (2013) do not allow for generalization to new observations, known as *calibration* (Aharoni et al., 2014). Therefore, they used the receiver operating characteristic (ROC) curve as a method to evaluate discrimination accuracy, and found that the ACC included in the prediction model resulted in a better prediction accuracy (and discrimination), than when ACC activity during the GNG task was not included as a predictor in the model, especially when predicting nonviolent crimes (i.e. AUC = 0.76 versus AUC = 0.68) compared to any crime (AUC = 0.68 versus AUC = 0.61). Note however, that these rearrest outcomes specifically looked at any sort of crime and nonviolent crime and does not give information on the predictive accuracy for violent crime.

Steele et al. (2015) added to the above studies by adding an EEG measurement in a subsample of Aharoni et al. (2013) and looked at the predictive accuracy of ERPs in addition to the earlier fMRI responses on the GNG task as a measure of impulse control and error processing difficulties. Steele and colleagues conducted different analyses and found that support vector machine neuroprediction models performed the best in predicting future rearrest by predicting 83.33 % of those rearrested correctly and that an increase in the Pe ERP amplitude in combination with a decrease in ACC activity during the GNG task (suggesting abnormal error-processing) resulted in the highest risk for recidivism. Surprisingly, the error-related negativity (ERN) ERP component was not a significant predictor future rearrest. The authors concluded that it is especially the post-error cognitive function such as conscious evaluation of the error, response strategy adjustments, and/or affective assessment of the error, that is associated with poor future behavioral outcomes.

Zijlmans et al. (2021) also used functional brain activity during the GNG task using fMRI, and the ERN and Pe as measures of error processing obtained during the Eriksen-Flanker task in predicting recidivism in 127 delinquent young adults. The authors found that neurobiological measures added to the predictive power above all other variables, and that the added predictive power decreased slightly over time. The model without neurobiological measures performed moderately well for both any offences and serious offences (i.e. violent offences), but adding neurobiological measures increased the predictive power. That is, while a model containing age, ethnicity, prior offences, cannabis use and self-reported aggressive behavior predicted serious offences with AUC = 0.75 accurately, the model containing those predictors + the neurobiological correlates performed with AUC = 0.80 accuracy. It was also found that while ACC activity assessed with fMRI was not predictive of recidivism, the ERN was predictive of any offense and the Pe was mostly predictive of serious offences.

Last, the outcomes presented by Delfin et al. (2019) concerned a sample of 44 forensic psychiatric patients that were followed-up for an average of ten years after discharge. The researchers were the first to study the added predictive value of resting-state regional cerebral blood flow, measured with single-photon emission computed tomography neuroimaging (SPECT). The results showed that their extended model including the resting-state regional cerebral blood flow measurements

performed better in predicting future risk than their baseline model with eight empirically established risk factors. More specifically, the AUC increased from 0.69 to 0.81, resulting in 12 out of 16 recidivists and 24 out of 28 non-recidivists being correctly classified by the model when neural measures were included compared with the model containing age at forensic institution, age at first crime, degree of psychopathic traits, sex, substance use disorder, cluster B personality disorder, educational attainment, mental disorder in first-degree relative, and previous criminality. Unfortunately, due to the low base-rate of the crimes in the follow-up period, no differentiation in types of crime, for instance violent crimes, could be made.

In sum, most of the prospective studies are conducted using functional neuroimaging indices and are conducted within the same sample of offenders. Also, most of the studies focused on neural measures that reflect cognitive processes of cognitive control, inhibition, and error monitoring, thus focusing on dysfunctional behavioral inhibition and learning from past mistakes and feedback. Neural regions that are found to be involved in these studies are the more frontal regions such as the ACC and parietal regions (Aharoni et al., 2013, 2014; Delfin et al., 2019).

It must be noted that the reviewed prospective/longitudinal studies did not look at other cognitive processes, such as emotional regulation and empathic responding, cognitive processes that are also found to be related to aggressive and violent behavior (e.g. Chialant et al., 2016; Maloney et al., 2023; Navas-Casado et al., 2023). In addition, in contrast to the cross-sectional studies, none of the prospective studies were aimed at neural connectivity measures as markers for (violent) recidivism.

4. Discussion

One of the major aims in forensic mental health is assessing future violence risk of forensic clients and subsequently managing this risk using various (treatment) interventions. However, as recent studies have shown that risk-assessment tools in forensic psychiatry are not so effective in predicting future risk for violence or recidivism as much as we would like to see in the field (see for review and meta-analysis Ogonah et al., 2023), there is an urgent need for innovation in risk assessment practices. At the same time, different scholars plea for a more holistic approach when it comes to risk assessment and treatment, pointing to a biopsychosocial model of looking at risk for violence (e.g., Cohn et al., 2016; Jansen, 2022; Van Dongen, 2020). Additionally, more and more studies in medicine, including psychiatry show the benefits of algorithms, using machine learning in predicting treatment outcomes (Feuerriegel et al., 2024). Therefore, the current study aimed to elucidate the potential of neurobiological correlates, more specifically structural and functional neuroimaging outcomes, to 'predict' (future) aggressive or violent (offending) behavior.

The current review included and discussed fourteen studies, of which the majority ($n = 8$) was cross-sectional, meaning that they did not 'predict' violence or recidivism at a later timepoint. We grouped the studies by whether they used a cross-sectional or longitudinal design and whether the data included structural or functional MRI indices. In addition, we were specifically interested to see if studies included indices of predictive accuracy. We found that the previous published papers included in our review were so divers in outcomes, that it was very difficult to give a general conclusion of the results. That is, although cross sectional studies found at times overlapping brain structures to be related to aggression or (violent) offending, such as dorsomedial structures (i.e. ACC), structures in the temporal pole (i.e. amygdala) and their connections, prospective studies did not. One of the reasons for these findings is, because the studies included region's of interest, they looked at structures that were already found to be related to particular antisocial tendencies and aggression.

Besides the fact that its difficult to infer clear conclusions regarding the brain regions associated with aggression or (violent) recidivism, only

a handful of studies specifically looked at the predictive accuracy of future recidivism. But although some of those studies did include more traditional risk predictors such as prior offending, substance use and/or social economic status, studies with a rigorous and systematic approach for studying the added accuracy in prediction using neuroimaging measures over and above traditional risk factors (for instance as measured with the HCR-20), are lacking. This is very important, because exactly that, the *added* value of neurobiology in the prediction of future violence, is of the main interest in those studies. Thus, for future research on this matter, it is of uttermost importance that those studies examine the incremental predictive accuracy and clinical utility of using neurobiological information in violence risk assessment. Those future studies thus need to include both traditional risk factors/assessment tools with additional neurobiological information to be able to validly conclude about the added value of neurobiology in violence risk assessment.

Importantly, 'neuroprediction' like other forms of prediction using AI algorithmic models such as machine learning, comes with ethical and legal challenges, including bias, stereotyping, 'discrimination', data privacy issues and the like (e.g., Tortora, 2024).

We also recognize that with this potential of neuro-imaging correlates as predictors, we also must outline other aspects and context in which this potential future new framework would work in forensic mental healthcare. That is, is it feasible to use neurobiological markers as 'predictors'? Additionally, do we in the field of forensic mental healthcare, want AI algorithms to take-over risk-assessment? In the next section, we will also discuss those questions.

4.1. Ethical and legal challenges

Since AI algorithms and 'neuroprediction' as a special form of these algorithms have risen in popularity over the years, scholars also increasingly acknowledge and raise the ethical and legal issues that seem inherently associated with these practices in forensic psychiatry.

Although we do not want to trivialize the potential threats that AI and neurotechnology can form in terms of ethics and the legal position of clients, we also think that we have to be careful not to see these innovations only in this light, as if these challenges only exist for these innovations. For instance, some scholars point to AI algorithms as being discriminative and foster prototypes leading to biased decision-making when using AI algorithms such as machine learning techniques in neuroprediction research (e.g., Tortora, 2024; Tortora et al., 2020). At the same time, it seems to be forgotten that the history of risk assessment and the development of risk assessment tools relies on using statistical models that also do exactly that. Risk assessment tools are developed based on group data that are used to classify groups of individuals into a either violent/non-violent, or recidivist or non-recidivist category based using statistical models such as linear regression, and predictive performance is determined using the Receiver Operating Characteristic Curve (the ROC curve) and subsequent Area Under the Curve (AUC) and related performance indicators (see Ogonah et al., 2023 for a review of studies).

Also, although AI and machine learning implies, or at least is sometimes portrayed as resulting in humans being only observers, being 'helpless' and 'victims' of neurotechnology, there are calls for new human rights to protect us from those new (neuro)technologies using 'neurorights' (i.e. Farahany, 2023, see also Yuste et al., 2017 and Ienca & Andorno, 2017), and we think legal issues *do* have to be taken into consideration, these discussions in our view also diminish the role the clinician can play in using these new technologies. For instance, in machine learning, one can also decide what data are entered in the models (i.e., supervised machine learning), and clinicians may also decide for themselves how they use that information. Risk assessment tools have also evolved from their use as actuarial tools to SPJs in which the clinician uses their clinical view to interpret assessment outcomes. On the other hand, this also may be considered hypocritical in the way

'neuropsychological' is judged as being discriminative and biased, while we cannot wave away biases that clinicians as simply 'humans' may have when interpreting assessment results and making decisions (see Escollà-Gascón et al., 2023; Neal et al., 2022). Therefore, claiming that AI algorithms are biased, when clinical judgments are similarly biased, may result in a false negative focus on the potential that neurotechnological innovations might bring.

One other ethical, or maybe more philosophical issue concerning AI algorithms and its use in 'neuropsychological' is the implication of the word 'prediction'. The word 'prediction' is found to imply that one seeks the best possible prediction model, no matter what data goes into the model. It might be the choice of breakfast one might have, or favorite color, these all can make good predictors when resulting in the most optimal prediction model. Although such a prediction model might then be viewed as being of clinical utility in terms of predictively accurate (i.e. when $AUC > 0.80$ for instance), it is of course at the same time also not of clinical utility. That is, although a perfect predictor, those predictors might not give any relevant information about which kind of situations a person might show violent behavior, nor does it give tools for deciding on which management, or treatment strategy might be useful. Therefore, neuropsychological in terms of AI generated algorithms in our view is a narrow view of 'prediction' and 'neuropsychological'. This is, something that is historically done in risk assessment already. Using historical items in risk assessment, such as previous convictions, is found to be one of the best predictors of future violent behavior. However, those risk factors are stable, and cannot be changed, and this it gives no information on how to change this risk factor for better outcomes (i.e. less risk for future violence). The same holds for risk factors such as socioeconomic status and aspects in upbringing such as abuse. We think that we need to seek 'predictors' that not only explain future risk the most optimally, but also informs us (i.e., the clinicians) what type of treatment possibilities may be suitable and effective. In terms of neuropsychological, studies in this review that used structural neuro-imaging techniques (e.g., Lett et al., 2018; Schiffer et al., 2017; Kiehl et al., 2018), address those more stable structures in the brain without addressing their functional consequences. Those functional studies (e.g., Aharoni et al., 2013, 2014; Zijlmans et al., 2021) explore possible dynamic neuropsychological, or neural risk factors that might be altered in future treatments, for example using non-invasive neuromodulation, or neurofeedback (e.g. Sergiou et al., 2022).

Given the above, we therefore propose for the development of neuropsychological informed structured professional judgement, that is, a form of risk assessment that is also informed by the neurobiology of an individual. We plea for a form of precision medicine, or personalized risk assessment based on not only historical, social, and psychological risk factors, but also neuropsychological information to optimize risk assessment and subsequent risk management, including (treatment) interventions. That way, we believe by looking at the neuropsychological needs, we can also increase the treatment response (i.e., responsivity) in that individual (forensic) client.

We would like to give an example. In forensic psychiatric settings, empathic abilities are found to be an important criminogenic need that needs to be addressed during treatment (Day et al., 2010). However, empathic abilities, especially affective empathy traits, are difficult to assess, and treatments are found not to be very successful in increasing such abilities (Vachon et al., 2014). Additionally, it is found that when empathic abilities are successfully improved during treatment, those are the cognitive aspects of empathy (such as mentalizing), and not the affective aspects of empathy (De Wied et al., 2020). Therefore, a neuropsychological way of assessing affective empathy, for instance using EEG (Cheng et al., 2014; van Dongen et al., 2018) might be promising, also because those indices or markers might subsequently be effectively targeted in treatments using neuromodulation (thus neuropsychological informed treatment; Sergiou et al., 2022).

4.2. Why would we want neuroscientific informed risk assessment?

Why would we want to use neuro-imaging based markers as risk factors for future violence and/or recidivism? Is it worth the trouble and the costs? One might argue that brain scans are only used for its 'sexiness' and it does not really aid in violence risk assessment practices. One argument often brought up is the fact that neurological scans are expensive. Yes, if looking at scans using MRI scanners and compared to psychological tests, the former are indeed far more expensive. But, when compared to EEG for instance, one cannot make that argument. EEG is relatively inexpensive and although it measures other neural indices than MRI, it is found to be promising in finding reliable and robust biomarkers (e.g. Férat et al., 2022; Klooster et al., 2023; Watts et al., 2022), that can subsequently be used to stratify between different treatments (e.g. Voetterl et al., 2023).

One might argue that neuroscience does not add value beyond (neuro)psychological assessment or behavioral data and that instead of using expensive brain scans (this only holds when referring to methods involving MRI, but not EEG) one might simply rely on behavioral data (e.g., Morse, 2022), but we think this is a narrow perspective. Of course, by using various kinds of neuropsychological tests, one can assess the degree of cognitive control or impulsivity, which can be included in risk assessment. These measures, however, do not say anything about where the problems in cognitive control might come from. Assessing these functions using psychological or neuropsychological tests does also not fully aid our understanding and ultimately, the advancement in treatment responsivity of those who we are assessing, in this case the forensic clients. As also discussed above, in other areas of mental healthcare, advances are being made in using neuro-imaging indices as biomarkers such as in the case of Alzheimer's disease and Mild Cognitive Impairment (e.g., Cecchetti et al., 2021; Jiao et al., 2023). These biomarkers also show predictive validity in treatment response for depression (Schwartzmann et al., 2023; Klooster et al., 2024). Nevertheless, further research remains crucial and its application to the individual level has yet to be established (although promising results have been found by Ball et al., 2017).

4.3. The future: towards neurobiological informed risk assessment and neurorehabilitation

Is this Utopia? Of course, neuroscientific informed risk assessment and management have a long road to go. We need to further study neural markers, their predictive accuracy, and efficacy in forensic mental health, and we sure need to address the different ethical and legal concerns that have been raised. On the other hand, as we outlined above, we are further than we think. Looking at other fields within medicine, precision medicine is found to be promising, and hints to possibilities of having a future in which we can use neuroscientific information and machine learning to make predictions at the individual level (Feuerriegel et al., 2024). Also, as discussed, although there are grounds for ethical concerns, some of them are no more of an issue than they currently are already and give thought for reconsidering forensic clinical practices as we conduct them now.

It is our opinion that this future might lie in using indices based on brain networks, something that has not been studied yet in the field of neuropsychological of violence risk. It is now being increasingly recognized that not individual brain regions, but functional brain networks and their properties are essential to individual differences in cognition and behavior (e.g., Bullmore & Sporns, 2009; Chen et al., 2022), including antisocial traits (e.g., Tillem et al., 2018; Tillem et al., 2023), and translating person-level brain network functioning to clinical practice has become the interest of new research (e.g., Ball et al., 2017). Therefore, future research might need to focus more on these brain networks and their topological properties as neural markers for violent behavior and recidivism.

Another area that we would like to address for future research is to

ultimately close the loop between neural markers based on brain topological properties and managing of violence risk through treatment of these dysfunctional properties to ultimately reduce violent behavior. That is, recent research not only demonstrated that neuromodulation techniques such as tDCS might be a successful avenue for reducing aggressive behavior in forensic psychiatric patients (Sergiou et al., 2022; Sergiou et al., 2023), but research has also found that topological properties can serve as biomarkers for treatment outcomes using neuromodulation (Hao et al., 2021), and can even be reshaped using neuromodulation (Vecchio et al., 2018). However, although promising, those studies are not conducted yet in forensic populations and therefore needs to be addressed in future research.

5. Conclusion

Although neuroscientific informed risk assessment studies reveal promising results, we have a long way to go in translating these neuroscientific findings into forensic clinical practice. Future research must build upon these results to support its efficiency and added value to current practices by validating and replicating findings. Moreover, we need to seriously think of how to implement such a new paradigm shift in forensic mental health risk assessment and management to foster these new innovative developments in forensic mental health.

CRediT authorship contribution statement

Josanne D.M. van Dongen: Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization. **Yudith Haveman:** Writing – review & editing, Project administration, Methodology. **Carmen S. Sergiou:** Writing – review & editing, Project administration, Methodology. **Olivia Choy:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

Data availability

Data will be made available on request.

References

- Abi-Dargham, A., & Horga, G. (2016). The search for imaging biomarkers in psychiatric disorders. *Nature Medicine*, 22(11), 1248–1255.
- Aharoni, E., Abdulla, S., Allen, C. H., & Nadelhoffer, T. (2022). 6 ethical implications of neurobiologically informed risk assessment for criminal justice decisions: A case for pragmatism.
- *Aharoni, E., Mallett, J., Vincent, G. M., Harenski, C. L., Calhoun, V. D., Sinnott-Armstrong, W., ... Kiehl, K. A. (2014). Predictive accuracy in the neuroprediction of rearrest. *Social Neuroscience*, 9(4), 332–336.
- *Aharoni, E., Vincent, G. M., Harenski, C. L., Calhoun, V. D., Sinnott-Armstrong, W., Gazzaniga, M. S., & Kiehl, K. A. (2013). Neuroprediction of future rearrest. *Proceedings of the National Academy of Sciences*, 110(15), 6223–6228.
- *Amaoui, S., Marín-Morales, A., Martín-Pérez, C., Pérez-García, M., Verdejo-Román, J., & Morawetz, C. (2023). Intrinsic neural network dynamics underlying the ability to down-regulate emotions in male perpetrators of intimate partner violence against women. *Brain Structure and Function*, 228(9), 2025–2040.
- Anderson, D. A. (2021). The aggregate cost of crime in the United States. *The Journal of Law and Economics*, 64(4), 857–885.
- Antonova, E., Sharma, T., Morris, R., & Kumari, V. (2004). The relationship between brain structure and neurocognition in schizophrenia: A selective review. *Schizophrenia Research*, 70(2–3), 117–145.
- Asaoka, Y., Won, M., Morita, T., Ishikawa, E., & Goto, Y. (2021). Comparable level of aggression between patients with behavioural addiction and healthy subjects. *Translational Psychiatry*, 11(1), 375.
- *Athanasios, M., Dumais, A., Tikasz, A., Lipp, O., Dubreucq, J. L., & Potvin, S. (2022). Increased cingulo-orbital connectivity is associated with violent behaviours in schizophrenia. *Journal of Psychiatric Research*, 147, 183–189.
- Ball, T. M., Goldstein-Piekarski, A. N., Gatt, J. M., & Williams, L. M. (2017). Quantifying person-level brain network functioning to facilitate clinical translation. *Translational Psychiatry*, 7(10), e1248.
- Barratt, E. S., Stanford, M. S., Kent, T. A., & Alan, F. (1997). Neuropsychological and cognitive psychophysiological substrates of impulsive aggression. *Biological Psychiatry*, 41(10), 1045–1061.
- Bauer, L. O., O'Connor, S., & Hesselbrock, V. M. (1994). Frontal P300 decrements in antisocial personality disorder. *Alcoholism: Clinical and Experimental Research*, 18(6), 1300–1305.
- Bogaerts, S., Spreen, M., Ter Horst, P., & Gerlsma, C. (2018). Predictive validity of the HKT-R risk assessment tool: Two and 5-year violent recidivism in a nationwide sample of Dutch forensic psychiatric patients. *International Journal of Offender Therapy and Comparative Criminology*, 62, 2259–2270.
- Borrebaeck, C. A. (2017). Precision diagnostics: Moving towards protein biomarker signatures of clinical utility in cancer. *Nature Reviews Cancer*, 17(3), 199–204.
- Brower, M. C., & Price, B. H. (2001). Neuropsychiatry of frontal lobe dysfunction in violent and criminal behaviour: A critical review. *Journal of Neurology, Neurosurgery, and Psychiatry*, 71, 720–726.
- Bullmore, E., & Sporns, O. (2009). Complex brain networks: Graph theoretical analysis of structural and functional systems. *Nature Reviews Neuroscience*, 10(3), 186–198.
- Calzada-Reyes, A., Alvarez-Amador, A., Galan-García, L., & Valdes-Sosa, M. (2013). EEG abnormalities in psychopath and non-psychopath violent offenders. *Journal of Forensic and Legal Medicine*, 20(1), 19–26.
- Carver, C. S., & Harmon-Jones, E. (2009). Anger is an approach-related affect: Evidence and implications. *Psychological Bulletin*, 135(2), 183–204. <https://doi.org/10.1037/a0013965>
- Casula, A., Milazzo, B. M., Martino, G., Sergi, A., Lucifora, C., Tomaiuolo, F., ... Vicario, C. M. (2023). Non-invasive brain stimulation for the modulation of aggressive behavior—A systematic review of randomized sham-controlled studies. *Life*, 13(5), 1220.
- Cecchetti, G., Agosta, F., Basaia, S., Cividini, C., Cursi, M., Santangelo, R., ... Filippi, M. (2021). Resting-state electroencephalographic biomarkers of Alzheimer's disease. *NeuroImage: Clinical*, 31, Article 102711.
- Chen, C. Y., Raine, A., Chou, K. H., Chen, I. Y., Hung, D., & Lin, C. P. (2016). Abnormal white matter integrity in rapists as indicated by diffusion tensor imaging. *BMC Neuroscience*, 17, 1–8.
- Chen, J., Tam, A., Kebets, V., Orban, C., Ooi, L. Q. R., Asplund, C. L., ... Yeo, B. T. (2022). Shared and unique brain network features predict cognitive, personality, and mental health scores in the ABCD study. *Nature Communications*, 13(1), 1–17.
- Cheng, Y., Chen, C., & Decety, J. (2014). An EEG/ERP investigation of the development of empathy in early and middle childhood. *Developmental Cognitive Neuroscience*, 10, 160–169.
- Chester, D. S., Lynam, D. R., Milich, R., & DeWall, C. N. (2017). Physical aggressiveness and gray matter deficits in ventromedial prefrontal cortex. *Cortex*, 97, 17–22.
- Chialant, D., Edersheim, J., & Price, B. H. (2016). The dialectic between empathy and violence: An opportunity for intervention? *The Journal of Neuropsychiatry and Clinical Neurosciences*, 28(4), 273–285.
- Cohn, M. D., Popma, A., Raine, A., & Cima, M. (2016). Biological approaches to externalising disorders and juvenile psychopathic traits. In *The handbook of forensic psychopathology and treatment* (pp. 74–96). Routledge.
- Convit, A., Czobor, P., & Volavka, J. (1991). Lateralized abnormality in the EEG of persistently violent psychiatric inpatient. *Journal of Biological Psychiatry*, 30, 363–370.
- Çorbacıoğlu, Ş. K., & Aksel, G. (2023). Receiver operating characteristic curve analysis in diagnostic accuracy studies: A guide to interpreting the area under the curve value. *Turkish Journal of Emergency Medicine*, 23(4), 195.
- De Wied, M., Van Der Graaff, J., De Rooij, G., Scheepers, F., Hoekstra, P. J., Branje, S., & Van De Schoot, R. (2020). The role of client empathy in treatment outcome in a sample of adolescents referred to forensic youth psychiatric services. *Children and Youth Services Review*, 118, Article 105301.
- Decety, J., Chen, C., Harenski, C. L., & Kiehl, K. A. (2015). Socioemotional processing of morally-laden behavior and their consequences on others in forensic psychopaths. *Human Brain Mapping*, 36(January), 2015–2026.
- Deco, G., Jirsa, V. K., & McIntosh, A. R. (2011). Emerging concepts for the dynamical organization of resting-state activity in the brain. *Nature Reviews Neuroscience*, 12(1), 43–56. <https://doi.org/10.1038/nrn2961>
- *Delfin, C., Krona, H., Andiné, P., Ryding, E., Wallinius, M., & Hofvander, B. (2019). Prediction of recidivism in a long-term follow-up of forensic psychiatric patients: Incremental effects of neuroimaging data. *PLoS One*, 14(5), Article e0217127.
- van Dongen, J. D., Brazil, I. A., van der Veen, F. M., & Franken, I. H. (2018). Electrophysiological correlates of empathic processing and its relation to psychopathic meanness. *Neuropsychology*, 32(8), 996.
- van Dongen, J. D., & Franken, I. H. (2019). Neuroscience in forensic psychiatry and psychology: An introduction to the special issue. *International Journal of Forensic Mental Health*, 18(3), 179–186.
- Douglas, K. S., Hart, S. D., Webster, C. D., Belfrage, H., de Vogel, V., de Vries Robbé, M., ... de Ruiter, C. (2013). *HCR-20 V3. Assessing Risk for Violence*.
- Ellingson, R. J. (1954). The incidence of EEG abnormality among patients with mental disorders of apparently nonorganic origin: A critical review. *American Journal of Psychiatry*, 111(4), 263–275.
- Escalà-Gascón, Á., Dagnall, N., & Drinkwater, K. (2023). Bias analysis in forensic and non-forensic psychiatric assessments. *International Journal of Law and Psychiatry*, 88, Article 101891.
- Farahany, N. A. (2023). *The battle for your brain: Defending the right to think freely in the age of neurotechnology*. St. Martin's Press.
- Fazel, S., Chang, Z., Fanshawe, T., Långström, N., Lichtenstein, P., Larsson, H., & Mallett, S. (2016). Prediction of violent reoffending on release from prison: Derivation and external validation of a scalable tool. *The Lancet Psychiatry*, 3(6), 535–543.

- Fazel, S., Fimińska, Z., Cocks, C., & Coid, J. (2016). Patient outcomes following discharge from secure psychiatric hospitals: Systematic review and meta-analysis. *British Journal of Psychiatry*, 208, 17–25.
- Féret, V., Arns, M., Deiber, M. P., Hasler, R., Perroud, N., Michel, C. M., & Ros, T. (2022). Electroencephalographic microstates as novel functional biomarkers for adult attention-deficit/hyperactivity disorder. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 7(8), 814–823.
- Feuerriegel, S., Frauen, D., Melnychuk, V., Schweisthal, J., Hess, K., Curth, A., ... van der Schaar, M. (2024). Causal machine learning for predicting treatment outcomes. *Nature Medicine*, 30(4), 958–968.
- Gao, Y., & Raine, A. (2009). P3 event-related potential impairments in antisocial and psychopathic individuals: A meta-analysis. *Biological Psychology*, 82(3), 199–210.
- Gatzke-Kopp, L. M., Raine, A., Buchsbaum, M., & LaCasse, L. (2001). Temporal lobe deficits in murderers: EEG findings undetected by PET. *The Journal of Neuropsychiatry and Clinical Neurosciences*, 13(4), 486–491.
- Gazzaniga, M. S. (Ed.). (2014). *Handbook of cognitive neuroscience*. Springer.
- van Goozen, S. H., Langley, K., & Hobson, C. W. (2022). Childhood antisocial behavior: A neurodevelopmental problem. *Annual Review of Psychology*, 73(1), 353–377.
- *Gou, N., Xiang, Y., Zhou, J., Zhang, S., Zhong, S., Lu, J., ... Wang, X. (2021). Identification of violent patients with schizophrenia using a hybrid machine learning approach at the individual level. *Psychiatry Research*, 306, Article 114294.
- Greicius, M. D., Krasnow, B., Reiss, A. L., & Menon, V. (2003). Functional connectivity in the resting brain: A network analysis of the default mode hypothesis. *Proceedings of the National Academy of Sciences of the United States of America*, 100, 253–258.
- Guy, L. S. (2008). Performance indicators of the structured professional judgment approach for assessing risk for violence to others: A meta-analytic survey.
- Hampel, H., Gao, P., Cummings, J., Toschi, N., Thompson, P. M., Hu, Y., ... Vergallo, A. (2023). The foundation and architecture of precision medicine in neurology and psychiatry. *Trends in Neurosciences*, 46(3), 176–198.
- Hanson, R. K. (1998). What do we know about sex offender risk assessment? *Psychology, Public Policy, and Law*, 4(1–2), 50.
- Hanson, R. K., & Morton-Bourgon, K. (2007). *The accuracy of recidivism risk assessments for sexual offenders: A meta-analysis*. Ottawa, ON: Public Safety and Emergency Preparedness Canada.
- Hanson, R. K., & Morton-Bourgon, K. E. (2009). The accuracy of recidivism risk assessments for sexual offenders: A meta-analysis of 118 prediction studies. *Psychological Assessment*, 21(1), 1.
- Hanson, R. K., & Thornton, D. (2000). Improving risk assessments for sex offenders: A comparison of three actuarial scales. *Law and Human Behavior*, 24, 119–136.
- Hao, J., Luo, W., Xie, Y., Feng, Y., Sun, W., Peng, W., ... Wang, X. (2021). Functional network alterations as markers for predicting the treatment outcome of cathodal transcranial direct current stimulation in focal epilepsy. *Frontiers in Human Neuroscience*, 15, Article 637071.
- Harmon-Jones, E. (2003). Clarifying the emotive functions of asymmetrical frontal cortical activity. *Psychophysiology*, 40(6), 838–848.
- Harris, G. T., Rice, M. E., & Quinsey, V. L. (1993). Violent recidivism of mentally disordered offenders: The development of a statistical prediction instrument. *Criminal Justice and Behavior*, 20, 315–335.
- Heilbrun, K., Leheny, C., Thomas, L., & Huneycutt, D. (1997). A national survey of US statutes on juvenile transfer: Implications for policy and practice. *Behavioral Sciences & the Law*, 15(2), 125–149.
- Hill, D., & Waterson, D. (1942). Electroencephalographic studies of psychopathic personalities. *Journal of Neurology, Neurosurgery, and Psychiatry*, 5, 47–65.
- Hofman, D., & Schutter, D. J. L. G. (2009). Inside the wire: Aggression and functional interhemispheric connectivity in the human brain. *Psychophysiology*, 46, 1054–1058. <https://doi.org/10.1111/j.1469-8986.2009.00849.x>
- Hofman, D., & Schutter, D. J. L. G. (2012). Asymmetrical frontal resting-state beta oscillations predict trait aggressive tendencies and behavioral inhibition. *Social Cognitive and Affective Neuroscience*, 7(7), 850–857. <https://doi.org/10.1093/scan/nrs060>
- Iacono, W. G., Malone, S. M., & McGue, M. (2003). Substance use disorders, externalizing psychopathology, and P300 event-related potential amplitude. *International Journal of Psychophysiology*, 48(2), 147–178.
- Ienca, M., & Andorno, R. (2017). Towards new human rights in the age of neuroscience and neurotechnology. *Life Sciences Society and Policy*, 13(1), 5.
- Jansen, L. M. (2022). The neurobiology of antisocial behavior in adolescence; current knowledge and relevance for youth forensic clinical practice. *Current Opinion in Psychology*, 47, Article 101356.
- Jiao, B., Li, R., Zhou, H., Qing, K., Liu, H., Pan, H., ... Shen, L. (2023). Neural biomarker diagnosis and prediction to mild cognitive impairment and Alzheimer's disease using EEG technology. *Alzheimer's Research & Therapy*, 15(1), 32.
- Kempes, M. (2023). Added value of neurotechnology for forensic psychiatric and psychological assessment. *Handbook of Clinical Neurology*, 197, 217–232.
- Keune, P., et al. (2012). Prefrontal brain asymmetry and aggression in imprisoned violent offenders. *Neuroscience Letters*, 151, 191–195. <https://doi.org/10.1016/j.neulet.2012.03.058> (PMID: 22484012).
- *Kiehl, K. A., Anderson, N. E., Aharoni, E., Maurer, J. M., Harenski, K. A., Rao, V., ... Steele, V. R. (2018). Age of gray matters: Neuroprediction of recidivism. *NeuroImage: Clinical*, 19, 813–823.
- Klooster, D., Voetterl, H., Baeken, C., & Arns, M. (2023). Evaluating robustness of brain stimulation biomarkers for depression: A systematic review of MRI and EEG studies. *Biological Psychiatry*, 95(6), 553–563.
- Klooster, D., Voetterl, H., Baeken, C., & Arns, M. (2024). Evaluating robustness of brain stimulation biomarkers for depression: A systematic review of magnetic resonance imaging and electroencephalography studies. *Biological Psychiatry*, 95(6), 553–563.
- Koh, L. L., Day, A., Klettke, B., Daffern, M., & Chu, C. M. (2020). The predictive validity of youth violence risk assessment tools: A systematic review. *Psychology, Crime & Law*, 26(8), 776–796.
- Lamsma, J., Mackay, C., & Fazel, S. (2017). Structural brain correlates of interpersonal violence: Systematic review and voxel-based meta-analysis of neuroimaging studies. *Psychiatry Research: Neuroimaging*, 267, 69–73.
- *Lett, T. A., Mohnke, S., Amelung, T., Brandl, E. J., Schiltz, K., Pohl, A., ... Walter, H. (2018). Multimodal neuroimaging measures and intelligence influence pedophile child sexual offense behavior. *European Neuropsychopharmacology*, 28(7), 818–827.
- Lindberg, N., Tani, P., Virkkunen, M., Porkka-Heiskanen, T., Appelberg, B., Naukkarinen, H., & Salmi, T. (2005). Quantitative electroencephalographic measures in homicidal men with antisocial personality disorder. *Psychiatry Research*, 136(1), 7–15.
- Maloney, M. A., Eckhardt, C. I., & Oesterle, D. W. (2023). Emotion regulation and intimate partner violence perpetration: A meta-analysis. *Clinical Psychology Review*, 100, Article 102238.
- Matthies, S., Rüsck, N., Weber, M., Lieb, K., Philipsen, A., Tiescher, O., ... van Elst, L. T. (2012). Small amygdala–high aggression? The role of the amygdala in modulating aggression in healthy subjects. *The World Journal of Biological Psychiatry*, 13(1), 75–81.
- Mednick, S. A., Volavka, J., Gabrielli, W. F., & Itil, T. M. (1981). EEG as a predictor of antisocial behavior. *Criminology*, 19, 219–229.
- Meynen, G., Van de Pol, N., Tesink, V., & Ligthart, S. (2023). Neurotechnology to reduce recidivism: Ethical and legal challenges. *Handbook of Clinical Neurology*, 197, 265–276.
- Molero-Chamizo, A., Riquel, R. M., Moriana, J. A., Nitsche, M. A., & Rivera-Urbina, G. N. (2019). Bilateral prefrontal cortex anodal tDCS effects on self-reported aggressiveness in imprisoned violent offenders. *Neuroscience*, 397, 31–40.
- Morse, S. J. (2022). Is executive function the universal acid? *Criminal Law and Philosophy*, 16(2), 299–318.
- Mossman, D. (1994). Assessing predictions of violence: Being accurate about accuracy. *Journal of Consulting Clinical Psychology*, 62, 783–792.
- Motzkin, J. C., Newman, J. P., Kiehl, K. A., & Koenigs, M. (2011). Reduced prefrontal connectivity in psychopathy. *Journal of Neuroscience*, 31, 7348–7357. <https://doi.org/10.1523/JNEUROSCI.4215-11.2011>. 22131397
- Moya-Albiol, L. (2004). Bases neurales de la Violencia. *Revista de Neurologia*, 38, 1067–1075.
- Nadelhoffer, T., Bibas, S., Grafton, S., Kiehl, K. A., Mansfield, A., Sinnott-Armstrong, W., & Gazzaniga, M. (2012). Neuroprediction, violence, and the law: Setting the stage. *Neuroethics*, 5, 67–99.
- Nadelhoffer, T., & Sinnott-Armstrong, W. (2012). Neurolaw and neuroprediction: Potential promises and perils. *Philosophy Compass*, 7(9), 631–642.
- Navas-Casado, M. L., García-Sánchez, E., & Salguero, J. M. (2023). Associations between maladaptive and adaptive emotion regulation strategies and aggressive behavior: A systematic review. *Aggression and Violent Behavior*, 71, Article 101845.
- Neal, T., Lienert, P., Denne, E., & Singh, J. P. (2022). A general model of cognitive bias in human judgment and systematic review specific to forensic mental health. *Law and Human Behavior*, 46(2), 99.
- Nikolic, M., Pezzoli, P., Jaworska, N., & Seto, M. C. (2022). Brain responses in aggression-prone individuals: A systematic review and meta-analysis of functional magnetic resonance imaging (fMRI) studies of anger and aggression-eliciting tasks. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 119, Article 110596.
- Ogonah, M. G., Seyedsalehi, A., Whiting, D., & Fazel, S. (2023). Violence risk assessment instruments in forensic psychiatric populations: A systematic review and meta-analysis. *The Lancet Psychiatry*, 10(10), 780–789.
- Page, M. J., Moher, D., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., ... McKenzie, J. E. (2021). PRISMA 2020 explanation and elaboration: Updated guidance and exemplars for reporting systematic reviews. *British Medical Journal*, 372. <https://doi.org/10.1136/bmj.n160>
- Pardini, D. A., Raine, A., Erickson, K., & Loeber, R. (2014). Lower amygdala volume in men is associated with childhood aggression, early psychopathic traits, and future violence. *Biological Psychiatry*, 75(1), 73–80.
- Patrick, C. J. (2008). Psychophysiological correlates of aggression and violence: an integrative review. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 363(1503), 2543–2555.
- Patrick, C. J., Iacono, W. G., Bernat, E., Malone, S. M., Krueger, R. F., & McGue, M. (2001). Brain potential correlates of the latent externalizing dimension of psychopathology. *Psychophysiology*, 38, S75.
- Peterson, C. K., Shackman, A. J., & Harmon-Jones, E. (2008). The role of asymmetrical frontal cortical activity in aggression. *Psychophysiology*, 45(1), 86–92.
- Poldrack, R. A., Monahan, J., Imrey, P. B., Reyna, V., Raichle, M. E., Faigman, D., & Buckholz, J. W. (2018). Predicting violent behavior: What can neuroscience add? *Trends in Cognitive Sciences*, 22(2), 111–123.
- Popma, A., & Raine, A. (2006). Will future forensic assessment be neurobiologic? *Child and Adolescent Psychiatric Clinics*, 15(2), 429–444.
- *Popovic, D., Wertz, M., Geisler, C., Kaufmann, J., Lähtenvuuo, M., Lieslehto, J., ... Schiltz, K. (2023). Patterns of risk—Using machine learning and structural neuroimaging to identify pedophilic offenders. *Frontiers in Psychiatry*, 14, Article 1001085.
- Puri, B. K., Counsell, S. J., Saeed, N., Bustos, M. G., Treasaden, I. H., & Bydder, G. M. (2008). Regional grey matter volumetric changes in forensic schizophrenia patients: An MRI study comparing the brain structure of patients who have seriously and violently offended with that of patients who have not. *BMC Psychiatry*, 8, 1–6.
- van der Put, C. E., Gubbels, J., & Assink, M. (2019). Predicting domestic violence: A meta-analysis on the predictive validity of risk assessment tools. *Aggression and Violent Behavior*, 47, 100–116.

- Quaresima, V., & Ferrari, M. (2019). Functional near-infrared spectroscopy (fNIRS) for assessing cerebral cortex function during human behavior in natural/social situations: A concise review. *Organizational Research Methods*, 22(1), 46–68.
- Raine, A. (2018). Antisocial personality as a neurodevelopmental disorder. *Annual Review of Clinical Psychology*, 14(1), 259–289.
- Raine, A., Venables, P. H., & Williams, M. (1990). Relationships between central and autonomic measures of arousal at age 15 years and criminality at age 24 years. *Archives of General Psychiatry*, 47(11), 1003–1007.
- Reyes, A. C., & Amador, A. A. (2009). Qualitative and quantitative EEG abnormalities in violent offenders with antisocial personality disorder. *Journal of Forensic and Legal Medicine*, 16(2), 59–63.
- Rosell, D. R., & Siever, L. J. (2015). The neurobiology of aggression and violence. *CNS Spectrums*, 20(3), 254–279.
- Day, A., Casey, S., & Gerace, A. (2010). Interventions to improve empathy awareness in sexual and violent offenders: Conceptual, empirical, and clinical issues. *Aggression and Violent Behavior*, 15(3), 201–208.
- de Ruigh, E. L., Bouwmeester, S., Popma, A., Vermeiren, R. R. J. M., van Domburgh, L., & Jansen, L. M. (2021). Using the biopsychosocial model for identifying subgroups of detained juveniles at different risk of re-offending in practice: A latent class regression analysis approach. *Child and Adolescent Psychiatry and Mental Health*, 15(1), 33.
- Scarpa, A., & Raine, A. (1997). Psychophysiology of anger and violent behaviour. *The Psychiatric Clinics of North America*, 20, 375–393.
- *Schiffer, B., Amelung, T., Pohl, A., Kaergel, C., Tenbergen, G., Gerwinn, H., ... Walter, H. (2017). Gray matter anomalies in pedophiles with and without a history of child sexual offending. *Translational Psychiatry*, 7(5), e1129.
- Schutter, D. J., de Weijer, A. D., Meuwese, J. D., Morgan, B., & van Honk, J. (2008). Interrelations between motivational stance, cortical excitability, and the frontal electroencephalogram asymmetry of emotion: A transcranial magnetic stimulation study. *Human Brain Mapping*, 29, 574–580.
- Schwartzmann, B., Quilty, L. C., Dhami, P., et al. (2023). Resting-state EEG delta and alpha power predict response to cognitive behavioral therapy in depression: A Canadian biomarker integration network for depression study. *Scientific Reports*, 13, Article 8418.
- Senior, M., Fazel, S., & Tsiachristas, A. (2020). The economic impact of violence perpetration in severe mental illness: A retrospective, prevalence-based analysis in England and Wales. *The Lancet Public Health*, 5, e99–106.
- Sergioli, C. S., Santarnecchi, E., Romanella, S. M., Wieser, M. J., Franken, I. H., Rassin, E. G., & van Dongen, J. D. M. (2022). Transcranial direct current stimulation targeting the ventromedial prefrontal cortex reduces reactive aggression and modulates electrophysiological responses in a forensic population. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 7(1), 95–107.
- Sergioli, C. S., Tatti, E., Romanella, S. M., Santarnecchi, E., Weidema, A. D., Rassin, E. G., ... van Dongen, J. D. (2023). The effect of HD-tDCS on brain oscillations and frontal synchronicity during resting-state EEG in violent offenders with a substance dependence. *International Journal of Clinical and Health Psychology*, 23(3), Article 100374.
- Sheehan, A. E., Bounoua, N., Miglin, R., Spielberg, J. M., & Sadeh, N. (2021). A multilevel examination of lifetime aggression: Integrating cortical thickness, personality pathology and trauma exposure. *Social Cognitive and Affective Neuroscience*, 16(7), 716–725.
- Singh, J. P., Desmarais, S. L., Hurdus, C., et al. (2014). International perspectives on the practical application of violence risk assessment: A global survey of 44 countries. *International Journal of Forensic Mental Health*, 13, 193–206.
- Singh, J. P., Grann, M., & Fazel, S. A. (2011). Comparative study of violence risk assessment tools: A systematic review and meta-regression analysis of 68 studies involving 25,980 participants. *Clinical Psychological Review*, 31, 499–513.
- Steadman, H. J., & Silver, E. (2000). Immediate precursors of violence among persons with mental illness: A return to a situational perspective. In *Violence among the mentally ill: Effective treatments and management strategies* (pp. 35–48).
- *Steele, V. R., Claus, E. D., Aharoni, E., Vincent, G. M., Calhoun, V. D., & Kiehl, K. A. (2015). Multimodal imaging measures predict rearrest. *Frontiers in Human Neuroscience*, 9, 425.
- Swets, J. A., Dawes, R. M., & Monahan, J. (2000). Better decisions through science. *Scientific American*, 283(4), 82–87.
- *Ter Harmsel, J. F., van Dongen, J. D., Zijlmans, J., van der Pol, T. M., Marhe, R., & Popma, A. (2022). Neurobiological responses towards stimuli depicting aggressive interactions in delinquent young adults and controls: No relation to reactive and proactive aggression. *Brain Sciences*, 12(2), 124.
- Tesink, V., Douglas, T., Forsberg, L., Lighthart, S., & Meynen, G. (2024). Right to mental integrity and neurotechnologies: Implications of the extended mind thesis. *Journal of Medical Ethics*. <https://doi.org/10.1136/jme-2023-109645>
- Tillem, S., Dotterer, H. L., Goetschius, L. G., Lopez-Duran, N., Mitchell, C., Monk, C. S., & Hyde, L. W. (2023). Antisocial behavior is associated with reduced frontoparietal network efficiency in youth. *Social Cognitive and Affective Neuroscience*, 18(1), Article nsad026.
- Tillem, S., Ryan, J., Wu, J., Crowley, M. J., Mayes, L. C., & Baskin-Sommers, A. (2016). Theta phase coherence in affective picture processing reveals dysfunctional sensory integration in psychopathic offenders. *Biological Psychology*, 119, 42–45.
- Tillem, S., van Dongen, J., Brazil, I. A., & Baskin-Sommers, A. (2018). Psychopathic traits are differentially associated with efficiency of neural communication. *Psychophysiology*, 55(9), Article e13194.
- To, W. T., De Ridder, D., Hart, J., Jr., & Vanneste, S. (2018). Changing brain networks through non-invasive neuromodulation. *Frontiers in Human Neuroscience*, 12, 128.
- Tortora, L. (2024). Beyond Discrimination: Generative AI Applications and Ethical Challenges in Forensic Psychiatry. *Frontiers in Psychiatry*, 15, 1346059.
- Tortora, L., Meynen, G., Bijlsma, J., Tronci, E., & Ferracuti, S. (2020). Neuroprediction and ai in forensic psychiatry and criminal justice: A neurolaw perspective. *Frontiers in Psychiatry*, 11, 220.
- Vachon, D. D., Lynam, D. R., & Johnson, J. A. (2014). The (non) relation between empathy and aggression: Surprising results from a meta-analysis. *Psychological Bulletin*, 140(3), 751.
- Van Dongen, J. D. (2020). The empathic brain of psychopaths: From social science to neuroscience in empathy. *Frontiers in Psychology*, 11, 695.
- Vecchio, F., Di Iorio, R., Miraglia, F., Granata, G., Romanello, R., Bramanti, P., & Rossini, P. M. (2018). Transcranial direct current stimulation generates a transient increase of small-world in brain connectivity: An EEG graph theoretical analysis. *Experimental Brain Research*, 236, 1117–1127.
- Voetterl, H. T., Sack, A. T., Olbrich, S., Stuiver, S., Rouwhorst, R., Prentice, A., ... Arns, M. (2023). Alpha peak frequency-based Brainmarker-I as a method to stratify to pharmacotherapy and brain stimulation treatments in depression. *Nature Mental Health*, 1(12), 1023–1032.
- Volavka, J. (1999). The neurobiology of violence: an update. *The Journal of Neuropsychiatry and Clinical Neurosciences*, 11(3), 307–314.
- Wang, Y., Zhao, Y., Qiu, J., Ybarra, O., Liu, L., & Huang, Y. (2012). Neural correlates of aggression among individuals from low and high socioeconomic status: An ERP study. *International Journal of Psychological Studies*, 4(4), 37.
- Watts, D., Pulice, R. F., Reilly, J., Brunoni, A. R., Kapczynski, F., & Passos, I. C. (2022). Predicting treatment response using EEG in major depressive disorder: A machine-learning meta-analysis. *Translational Psychiatry*, 12(1), 332.
- Webster, C. D., Douglas, K. S., Eaves, D., & Hart, S. D. (1997). *HCR-20: Assessing risk for violence, version 2*. Burnaby, Canada: Mental Health, Law, and Policy Institute, Simon Fraser University.
- Widmayer, S., Sowislo, J. F., Jungfer, H. A., Borgwardt, S., Lang, U. E., Stieglitz, R. D., & Huber, C. G. (2018). Structural magnetic resonance imaging correlates of aggression in psychosis: A systematic review and effect size analysis. *Frontiers in Psychiatry*, 9, 217.
- Wong, T. Y., Sid, A., Wensing, T., Eickhoff, S. B., Habel, U., Gur, R. C., & Nickl-Jockschat, T. (2019). Neural networks of aggression: ALE meta-analyses on trait and elicited aggression. *Brain Structure and Function*, 224, 133–148.
- Yang, Y., & Raine, A. (2009). Prefrontal structural and functional brain imaging findings in antisocial, violent, and psychopathic individuals: A meta-analysis. *Psychiatry Research: Neuroimaging*, 174(2), 81–88.
- *Yu, T., Pei, W., Xu, C., Zhang, X., & Deng, C. (2022). Prediction of violence in male schizophrenia using sMRI, based on machine learning algorithms. *BMC Psychiatry*, 22(1), 676.
- Yuste, R., Goering, S., Agüera y Arcas, B., et al. (2017). Four ethical priorities for neurotechnologies and AI. *Nature*, 551, 159.
- Zijlmans, J., Marhe, R., Bevaart, F., Van Duin, L., Luijckx, M. J. A., Franken, I., ... Popma, A. (2021). The predictive value of neurobiological measures for recidivism in delinquent male young adults. *Journal of Psychiatry and Neuroscience*, 46(2), E271–E280.